CASE STUDY 02

Progressive Electricity Billing

1. PROBLEM STATEMENT

Calculate household electricity bills using consumption slabs, a fixed charge, and a surcharge. Compare bills for multiple

months and identify an unusual increase.

2. OBJECTIVES

• Calculate energy charges using progressive slabs.

• Add fixed charges and a surcharge when applicable.

• Compare two monthly bills and flag a large increase.

3. ASSUMPTIONS

• First 100 units cost ■2 per unit.

• Next 100 units cost ■3 per unit.

• Units above 200 cost ■5 per unit.

• Fixed charge is ■50. A 5% surcharge applies when consumption exceeds 250 units.

• An increase greater than 30% compared with the previous month is unusual.

4. ALGORITHM

1. Read previous-month and current-month units.

2. Validate that each unit value is not negative.

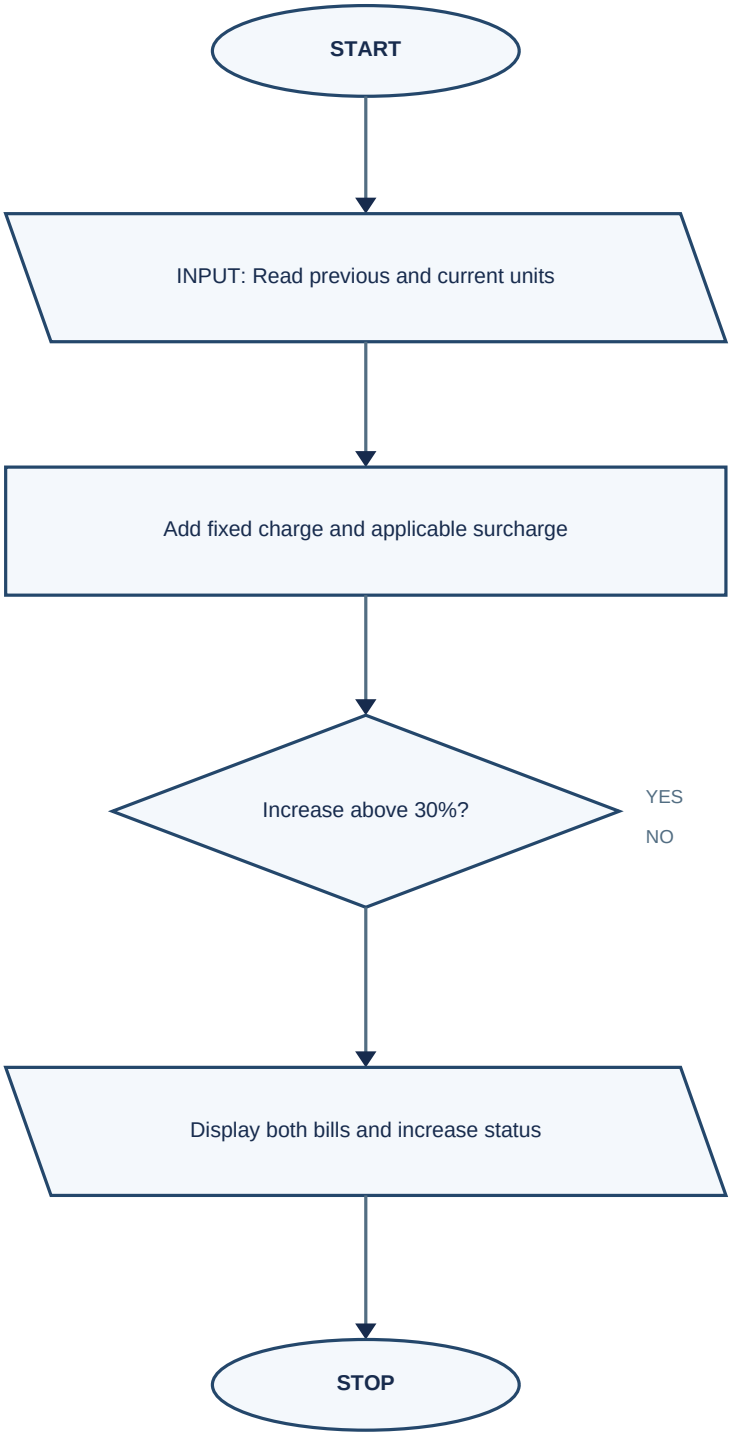
3. Calculate each bill using the progressive energy slabs and add the fixed charge.

4. Apply the 5% surcharge when usage exceeds 250 units.

5. Calculate percentage increase and display whether it exceeds 30%.

5. FLOWCHART

Oval: START/STOP | Parallelogram: input/output | Rectangle: process/calculation | Diamond: decision



6. PSEUDOCODE

```
BEGIN
INPUT previous_units, current_units
FOR each monthly usage
IF units < 0: RETURN invalid
CALCULATE energy charge using progressive slabs
ADD fixed charge
IF units > 250: ADD 5% surcharge
CALCULATE previous_bill and current_bill
CALCULATE percentage increase
IF increase > 30: DISPLAY unusual increase
ELSE: DISPLAY increase within limit
END
```

7. PYTHON SOURCE CODE

```
def calculate_bill(units):
if units < 0:
return -1
if units <= 100:
energy = units * 2
elif units <= 200:
energy = 100 * 2 + (units - 100) * 3
else:
energy = 100 * 2 + 100 * 3 + (units - 200) * 5
bill = energy + 50
if units > 250:
bill = bill + bill * 0.05
return bill
previous_units = 150
current_units = 280
previous_bill = calculate_bill(previous_units)
current_bill = calculate_bill(current_units)
print("Previous bill: Rs.", round(previous_bill, 2))
print("Current bill: Rs.", round(current_bill, 2))
increase = ((current_bill - previous_bill) / previous_bill) * 100
print("Increase:", round(increase, 2), "%")
if increase > 30:
print("Unusual increase detected")
else:
print("Increase is within the limit")
```

8. TEST CASES & EXPECTED RESULTS

TEST INPUT / CONDITION REASONING EXPECTED RESULT

100 units 100 × ■2 + ■50 ■250

150 units ■200 + 50 × ■3 + ■50 ■400

280 units Energy ■900 + fixed ■50, then 5% surcharge ■997.50 (including 5% surcharge)

9. SAMPLE OUTPUT

```
Previous bill: Rs. 400
Current bill: Rs. 997.5
Increase: 149.38 %
Unusual increase detected
```

10. CONCLUSION

The program applies progressive slab rates correctly and helps identify a sudden increase in household

electricity

expenses.

SYLLABUS CONCEPTS Variables • Data Types • Operators • Conditions • Loops • Lists • Nested Lists • Tuples • String Operations